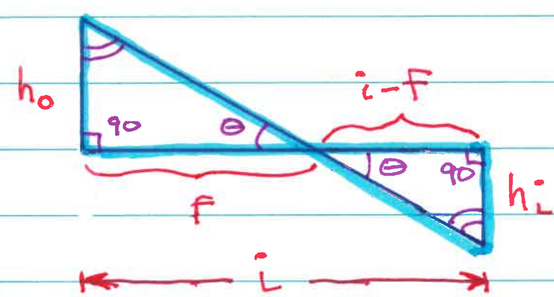
Lecture 5

1/30/26
JRA

Homework Hints

Discussed two homework problems from homework assignment due today

- **Problem 3.7** Derivation of Thin Lens Equation



$\tan\theta = \frac{h_o}{f}$ Upper triangle

$\tan\theta = \frac{h_i}{i-f}$ Lower triangle

$$\frac{h_o}{f} = \frac{h_i}{i-f} \Rightarrow \boxed{\frac{h_o}{h_i} = \frac{f}{i-f}}$$

Repeat process for yellow triangles in book

This problem can be solved using similar triangles OR the definition of tangent

- **Problem 3.16** Two-Lens System

(a) Note: f is negative for a diverging lens

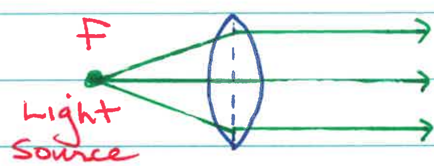
(b) When image from 1st lens is object for 2nd lens, compute distance to 2nd

Here, for diverging object is $48-30 = 18$ cm away

(c) Look at ray tracing tutorial & sign conventions

Homogeneous Illumination

To obtain homogeneous light, put light source at focal point



Can use same diagram going either way.
Confirm w/ Snell's Law.

See Demo in last lecture.

Image Quality

Image quality is a pervasive issue w/ all imaging techniques, so we'll revisit it again & again.

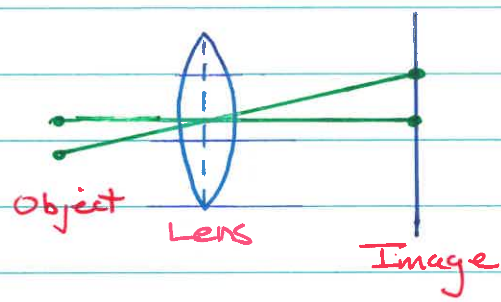
One key measure of quality is "resolution."

The determinants of resolution vary w/ technique, but diffraction appears over & over as a major determinant.

What is resolution? Subtle.

It's standard in μscopy to use a one-parameter descriptor of resolution.

Ray optics – NO DIFFRACTION (because it's ignored)
⇒ NO resolution problem



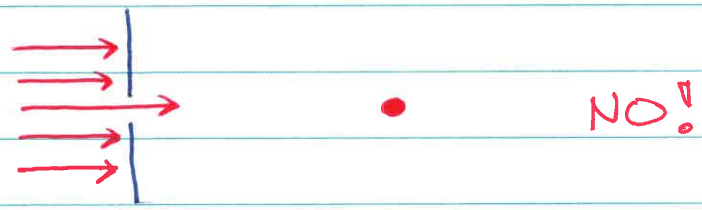
Two points in object map into two points in image, in the absence of diffraction

Resolution – can you tell that two points in the object are still two points when you look at the image?

Ray optics is an approximation: need wave optics to understand resolution. This will come up w/ ultrasound too.

Diffraction is wave optics.

Light & sound are waves. Can see & hear around corners, apertures, etc. because of diffraction. This applies to openings & obstacles.



Light spreads at hole, so image downstream is NOT just a bright spot.

Little aperture ⇒ MORE spreading

Big aperture ⇒ LESS spreading

Key Facts about spreading

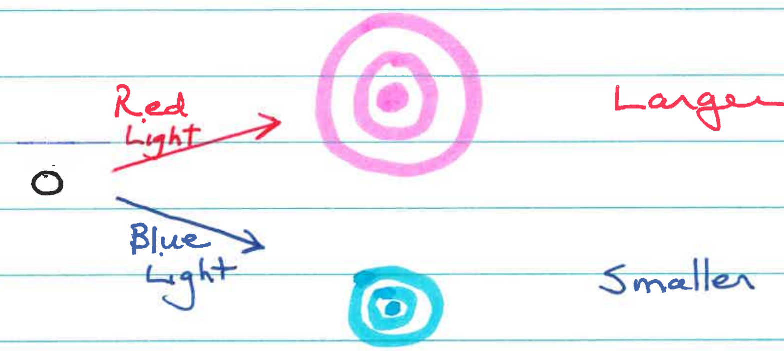
(1) Spreading increases as aperture size decreases

(2) Spreading increases as wavelength λ increases

Red light ⇒ worse
Blue light ⇒ better

$\boxed{\text{ANIMATION}}$ PhET Diffraction

Used simulation to examine circular apertures because they correspond to most biological situations



Aperture Size - looked at how aperture size affects the size of the diffraction pattern

Smaller aperture ∘ ⇒ Larger pattern
Larger aperture O ⇒ Smaller pattern

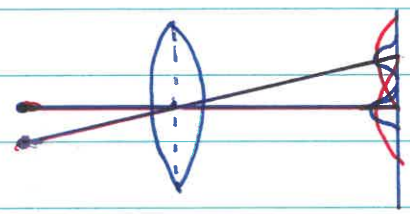
Wavelength - looked at how wavelength of light affects the size of the diffraction pattern

Longer wavelength (red) ⇒ Larger pattern
Shorter wavelength (blue) ⇒ Smaller pattern

Looked especially at size of central maximum, which comprises about 80% of the total intensity

Geometrical (ray) optics predicts that a point will be imaged as a point

Wave optics predicts spreading due to diffraction ⇒ a point is NOT imaged as a point



Blue → narrow spots
Red → wide spots

Here, blue light patterns are narrow relative to spacing ⇒ resolvable despite spacing

Red light patterns are wide & overlap significantly ⇒ NOT resolvable due to blurring

Big qualitative take-home messages:

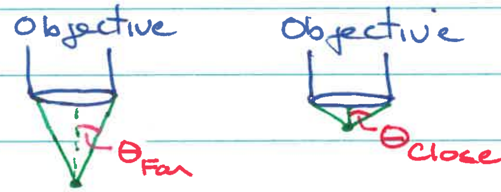
(1) Want narrow diffraction patterns so closely spaced objects can be resolved

(2) Resolution improves if aperture (e.g., lens) is large ⇒ use big lenses AND if wavelength is short ⇒ use blue light

Use < Big lenses / Short wavelengths

Want to gather as much light as possible

$\boxed{\text{ANIMATION}}$ MoI Expressions Numerical Aperture



$\theta_{Close} > \theta_{Far}$

⇒ Closer lens captures a larger cone of light

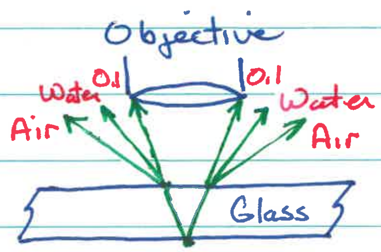
This is why you usually work with microscope objective very close to specimen

"Numerical Aperture" can be used to quantify the amount of light capture

$\boxed{\text{ANIMATION}}$ MoI Expressions NA & Resolution

Higher NA ⇒ better resolution (narrower diffraction pattern)

$\boxed{\text{ANIMATION}}$ MoI Expressions Immersion Oil & Refractive Index



Rays bend away from the objective when exiting the glass if the medium between the glass & objective consists of air or water ⇒ NOT captured

Rays do <u>NOT</u> bend when oil is the intervening medium

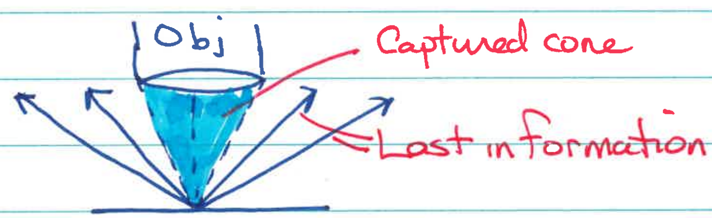
Better capture: Oil > Water > Air

Specimens emit light.

The more light you capture the more information you get & the less distortion the image suffers

⇒ Gather as much light as possible from the specimen

Do <u>NOT</u> want to throw light away.



Want captured cone to have a large capture angle

- <u>Separation</u> - specimen should be close to lens

- Intervening Medium

Oil or water between specimen & lens is helpful.
↑Best ↑Good

Oil & water help because they reduce light lost to refraction (bending)

- Wavelength - shorter λs are better